

4.3 Gestalt principles ↕

□ Watch: Data visualisation and design – part 2 [17.20 mins]

Watch the following video, which explains the four main Gestalt principles and how they can be applied. The advantages of refining the data-ink ratio and improving the effectiveness of using tables as visualisations are also explained.

This is useful as it provides an understanding of how people interpret what they see.



[Transcript](#)

(<https://learn.usc.edu.au/courses/35242/files/3138641?wrap=1>).

Source: © The University of Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Data visualisation and design – part 2](https://learn.usc.edu.au/courses/35242/files/3138660?wrap=1) (<https://learn.usc.edu.au/courses/35242/files/3138660?wrap=1>)

The Gestalt principles describe how people interpret and perceive objects. That is, how they put order and meaning into the things that they see. There are four principles that are important when designing data visualisations.

- *Similarity* relates to the fact that *objects with similar characteristics* are usually considered as belonging together (to the same set or group).

- *Proximity* refers to the fact that *objects that are physically close together* are usually seen as belonging together.
- *Enclosure* refers to the fact that *objects that are in the same enclosed area* are usually considered as belonging together.
- *Connection* refers to the fact that *objects that are connected in some manner* are usually considered as belonging together.

□ Watch: Gestalt psychology and why it's essential for good design [2.00 mins] and The Gestalt principles [10.41 mins]

These two videos are useful for understanding what helps our brains to perceive what we see. An explanation of Gestalt psychology is provided. Gestalt psychology assumes that our brains create a perception that is greater than the individual sensory inputs. The Gestalt principles are also described and illustrated.

As you watch the videos, think of a visualisation you have seen in your work environment. After you have finished watching the videos, reflect and make notes on:

- How you would have approached using the Gestalt principles in the data visualisation?
- What techniques could have been used, and what is the rationale behind using those techniques?



© HubSpot Marketing (2017).



© Daniel Storage (2019).

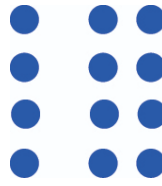
☐ Activity: Check your understanding of the Gestalt principles

Consider the icons presented in the cards below and drag them next to the Gestalt principle it best represents.

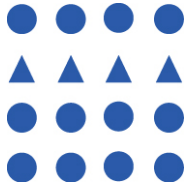
You have made 4 attempts. You got 100% correct on your last attempt.

**Good Figure**

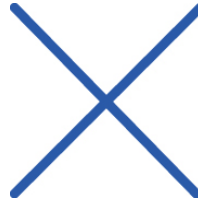
Objects grouped together tend to be perceived as a single figure. Tendency to simplify.

**Proximity**

Objects tend to be grouped together if they are close to each other.

**Similarity**

Objects tend to be grouped together if they are similar.

**Continuation**

When there is an intersection between two or more objects, people tend to perceive each object as a single uninterrupted object.

**Closure**

Visual connection or continuity between sets of elements which do not actually touch each other in a composition.

**Symmetry**

The object to be perceived as symmetrical shapes that form around their center.



★ 6/6

Source: ©ux hints (n.d.)